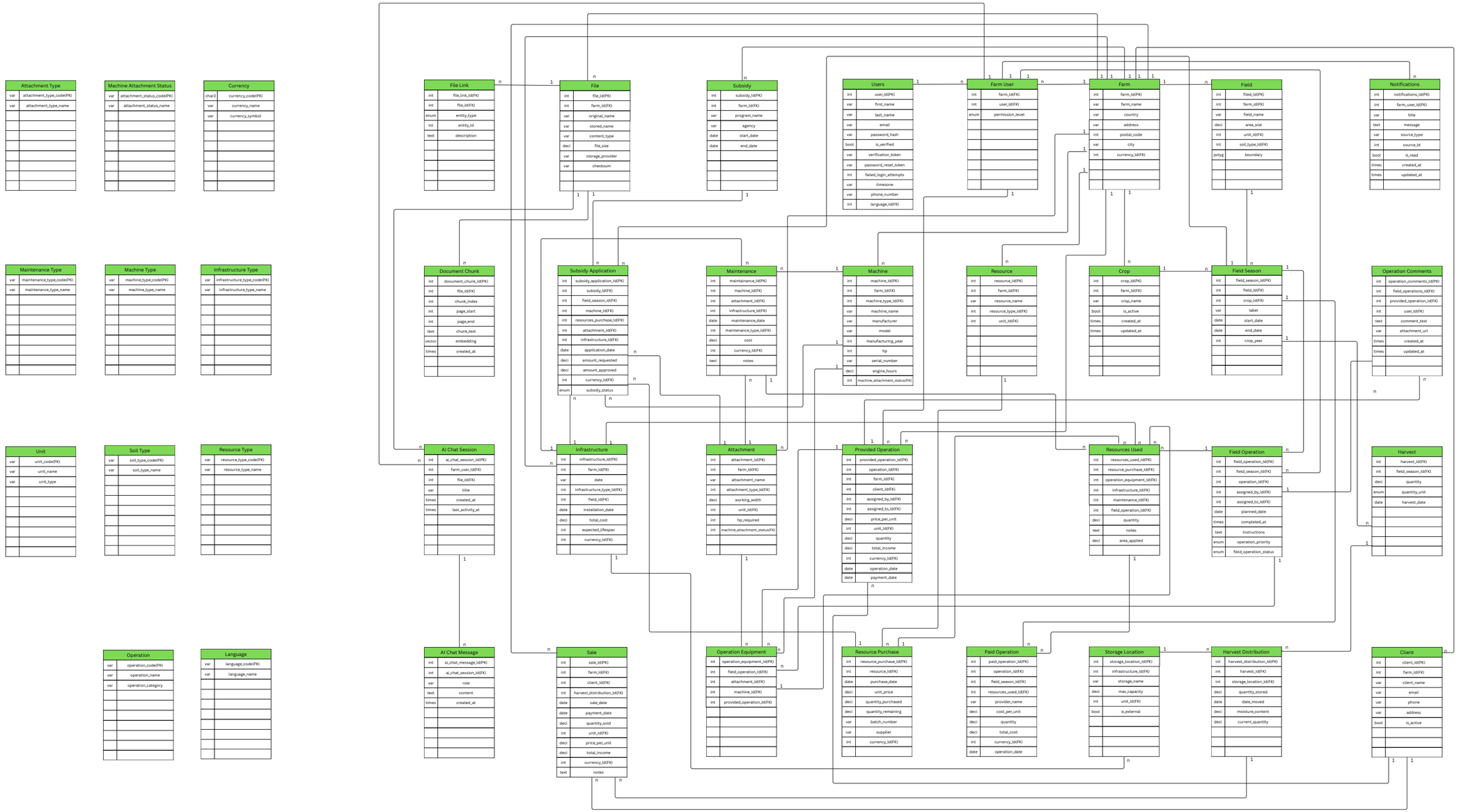Lookup Tables